

month return $r_{t-h}^s/\sigma_{t-h-1}^s$ and lagged sign of return, $sign(r_{t-h}^s)$. The equations are specified as follows:

$$r_t^s/\sigma_{t-1}^s = \alpha + \beta_h r_{t-h}^s/\sigma_{t-h-1}^s + \epsilon_t^s, \quad (7)$$

$$r_t^s/\sigma_{t-1}^s = \alpha + \beta_h sign(r_{t-h}^s) + \epsilon_t^s, \quad (8)$$

where r_t^s is the excess return of asset s in month t adjusted by its available ex ante volatility σ_{t-1}^s . $sign(r_{t-h}^s)$ takes the value $+1$ if $r_{t-h}^s \geq 0$ or -1 if $r_{t-h}^s < 0$. h is the number of lags used in the regressions which ranges from 1 to 60. Finally, ϵ_t^s denotes the error term, which has zero mean and finite variance.

Second, we repeat the above regressions by employing two alternative regressors: (i) the 12-month period return $PR_{t-h-11,t-h}$, and (ii) the 12-month mean return $sign P_{t-h-11,t-h}$. These two variables are directly related to the formation of TSM and RSM signals. Finally, the TSM and RSM signals, namely $S_{t-h-11,t-h}^{TSM}$ and $S_{t-h-11,t-h}^{RSM}$, are used as the third set of regressors. TSM and RSM signals take the value of $+1$ when $PR_{t-h-11,t-h}$ and $P_{t-h-11,t-h}$ indicates a buy signal, and the value of -1 for a sell signal.

In order to avoid the return co-movement across different futures contracts and keep the regression results adjusted for time series dependence, we calculate the t-statistics based on the robust standard error method proposed by [Thompson \(2011\)](#). This approach addresses the problem that the residuals are correlated across both firms (in our case, different futures contracts) and time.¹⁰ Therefore, it generates consistent asymptotic efficiency standard errors and perform relatively well in small sample situations.

Figure 2 shows the time series predictability of the previously mentioned three sets of explanatory variables on future returns. In Panel A, we can see that it is not

¹⁰[Moskowitz et al. \(2012\)](#) claimed that their OLS pooled regression using volatility scaled return is equivalent to a GLS approach. This means that they did use standard errors that cluster by both futures contract and time effect. However, they did not adjust for simultaneous correlation across both futures contract and time.